

Summary

Software engineer with 4 years at AWS building backend systems for large-scale sales planning workflows. Experienced in distributed processing, data-intensive services, system design, and developer tooling, with a focus on scalable and reliable production systems.

Skills

Languages: TypeScript, JavaScript, Java, Python, Go, SQL

Technologies: AWS, DynamoDB, Lambda, SQS, OpenSearch, API Gateway, MCP, AgentCore, PostgreSQL, Node.js, React, GraphQL

Experience

Amazon Web Services (AWS) / Software Development Engineer II / Seattle, Washington

Sep 2022 - Present

- Built backend systems and developer tooling that support **AWS sales allocation and quota planning**, including large-scale customer updates, validation, and planning workflows.
- Automated **94K+ customer updates** between consecutive sales-planning cycles, replacing a manual recreation and submission process. Built a fault-tolerant pipeline with retryable stages, persistent state, consistency checks, monitoring, and redrive tooling.
- Scaled bulk customer updates from **~1K to 100K records**. Replaced upfront resolution with checkpointed streaming jobs and redesigned DynamoDB partitioning and progress tracking to eliminate hot partitions and write contention.
- Built a rule-evaluation system that precomputes valid customer attributes for each sales territory. Used dependency-ordered parallel processing and OpenSearch indexing to power low-latency suggestions and backend validation.
- Designed authentication and API integration for an MCP gateway supporting IAM/SigV4 system agents and JWT-authenticated end users. Routed AgentCore through API Gateway to reuse existing Smithy-generated contracts and Lambda infrastructure.
- Co-designed a scheduler for concurrent customer-change transactions using a **disjoint-set data structure** to group overlapping operations. Dependent changes run serially while independent groups execute concurrently, preserving valid customer state.

Highlight / Software Developer Contractor / Remote

Jan 2022 - May 2022

- Built product and backend functionality for an open-source observability platform using **Go, GraphQL, PostgreSQL, React, and TypeScript**.

Amazon / Software Development Engineer Intern, Fire OS Multimedia / Sunnyvale, California

Summer 2021

- Developed a camera driver enabling wireless cameras to work with third-party applications on Fire TV devices.

Nepris (now Pathful) / Software Developer Contractor / Frisco, Texas

Jun 2019 - May 2021

- Migrated a legacy .NET/JSP application toward **React and Next.js**, building routing infrastructure that allowed legacy and modern experiences to coexist during the transition.

Audible / Software Development Engineer Intern / Newark, New Jersey

Summers 2019 & 2020

- Built **Java/Spring** services for audiobook recommendations on Amazon.com and an Alexa skill using Lambda, Node.js, and TypeScript.

Projects

Graded

- Created and scaled an iOS/Android grade-tracking application to **35K+ daily active users** across Texas school districts, helping students track and analyze their grades. **Acquired by GradePro in 2018.**

zig-sokol-crossplatform-starter

- Built a cross-platform **Zig/Sokol** application template targeting iOS, Android, Windows, and macOS from one build system. Added Android App Bundle generation, Xcode project generation, and GitHub Actions CI/CD.

KOTH

- Built an automated code-judging platform used for the TAMU Datathon 2020 Stock Challenge. Queued workers executed submissions in isolated Docker containers with customizable evaluators, backed by a web submission and leaderboard experience.

Codescrim

- Created an online platform for computer science clubs and classrooms to run automated coding competitions, manage problem sets, and track group leaderboards; used it to host Codecember, a coding competition I organized.

Education

Texas A&M University / College Station, Texas

- B.S. Computer Science, Minor in Business